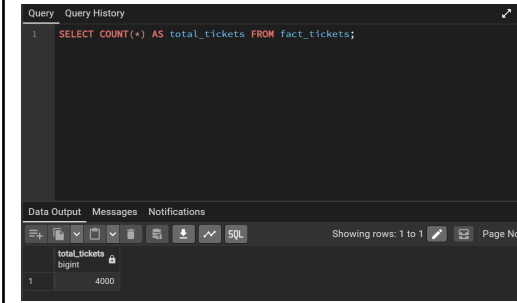
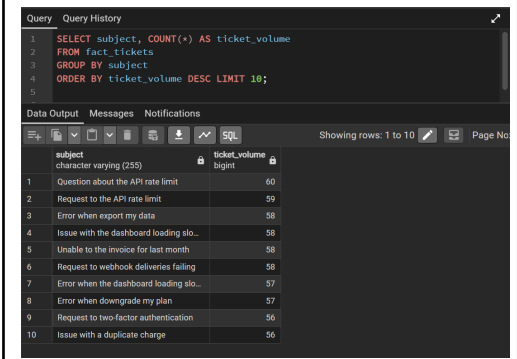
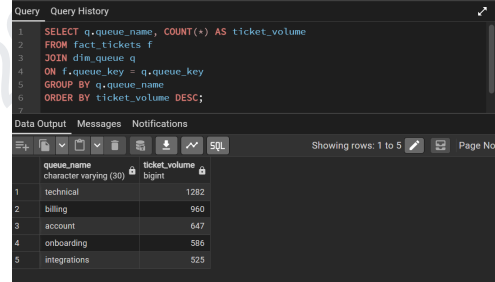
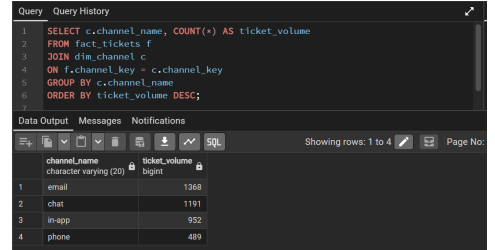
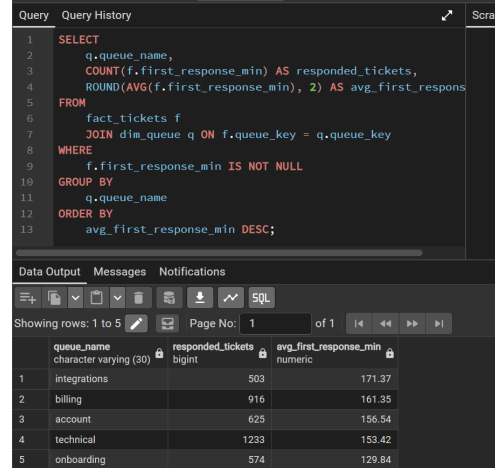


CUSTOMER SUPPORT TICKET ANALYTICS

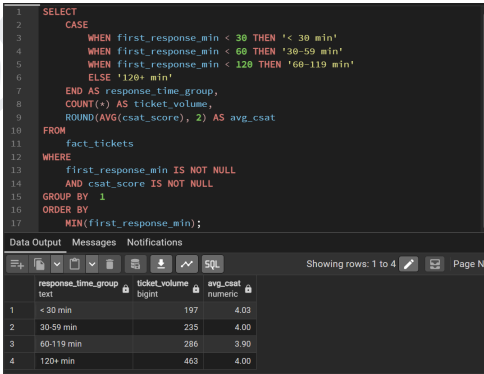
SQL Analysis

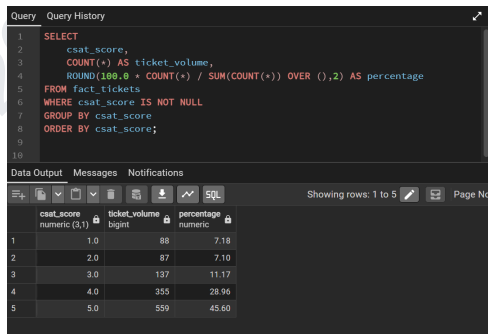
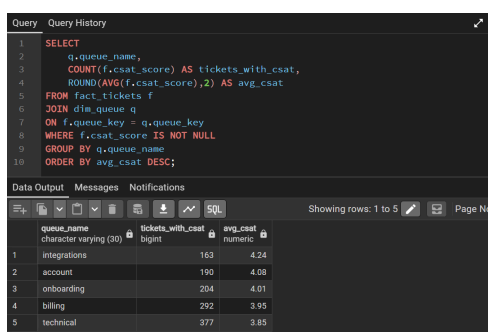
Query	Business Question / Requirement	BRD Reference	Analysis Purpose	SQL Query	Result
Q1 – Overall Ticket Volume	What is the total ticket volume?	FR-01	Measure the total number of tickets in the dataset.	SELECT COUNT(*) AS total_tickets FROM fact_tickets;	
Q2 – Top Ticket Subjects	Which issues (subjects) generate the most tickets?	BQ-01, FR-03	Identify the most frequently reported ticket subjects.	SELECT subject, COUNT(*) AS ticket_volume FROM fact_tickets GROUP BY subject ORDER BY ticket_volume DESC LIMIT 10;	

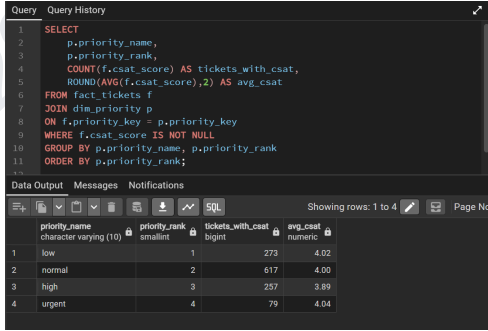
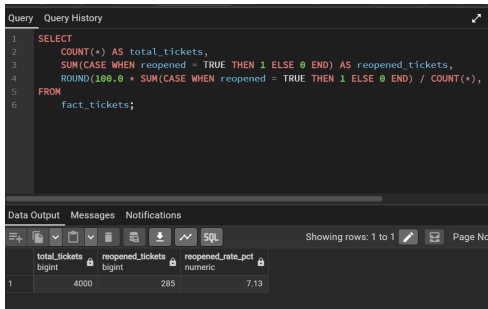
Q3 – Ticket Volume by Queue	Which queues have the highest workload?	BQ-02, FR-02	Measure workload distribution across support queues.	<pre>SELECT q.queue_name, COUNT(*) AS ticket_volume FROM fact_tickets f JOIN dim_queue q ON f.queue_key = q.queue_key GROUP BY q.queue_name ORDER BY ticket_volume DESC;</pre>	
Q4 – Ticket Volume by Channel	Which channels generate the most tickets?	BQ-03, FR-02	Measure customer support demand across channels.	<pre>SELECT c.channel_name, COUNT(*) AS ticket_volume FROM fact_tickets f JOIN dim_channel c ON f.channel_key = c.channel_key GROUP BY c.channel_name ORDER BY ticket_volume DESC;</pre>	
Q5 – First Response by Queue	Which queues have slower first-response times?	BQ-04, FR-04	Compare average first-response time across queues.	<pre>SELECT q.queue_name, COUNT(f.first_response_min) AS responded_tickets, ROUND(AVG(f.first_response_min), 2) AS avg_first_response_min FROM fact_tickets f JOIN dim_queue q ON f.queue_key = q.queue_key WHERE f.first_response_min IS NOT NULL GROUP BY q.queue_name ORDER BY avg_first_response_min DESC;</pre>	

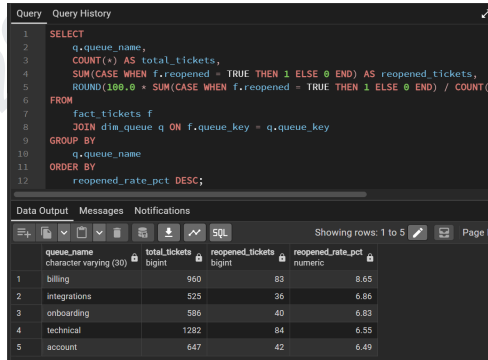
Q6 – First Response by Priority	Which priorities have slower first-response times?	FR-04	Compare average first-response time across priority levels.	<pre>SELECT p.priority_name, p.priority_rank, COUNT(f.first_response_min) AS responded_tickets, ROUND(AVG(f.first_response_min), 2) AS avg_first_response_min FROM fact_tickets f JOIN dim_priority p ON f.priority_key = p.priority_key WHERE f.first_response_min IS NOT NULL GROUP BY p.priority_name, p.priority_rank ORDER BY p.priority_rank;</pre>	Query Query History <pre>1 SELECT 2 p.priority_name, 3 p.priority_rank, 4 COUNT(f.first_response_min) AS responded_tickets, 5 ROUND(AVG(f.first_response_min), 2) AS avg_first_respons 6 FROM 7 fact_tickets f 8 JOIN dim_priority p ON f.priority_key = p.priority_key 9 WHERE 10 f.first_response_min IS NOT NULL 11 GROUP BY 12 p.priority_name, 13 p.priority_rank 14 ORDER BY 15 p.priority_rank;</pre> Data Output Messages Notifications Showing rows: 1 to 4 Page No: 1 of 1 <table><thead><tr><th></th><th>priority_name character varying (10)</th><th>priority_rank smallint</th><th>responded_tickets bigint</th><th>avg_first_response_min numeric</th></tr></thead><tbody><tr><td>1</td><td>low</td><td>1</td><td>828</td><td>302.62</td></tr><tr><td>2</td><td>normal</td><td>2</td><td>1857</td><td>149.50</td></tr><tr><td>3</td><td>high</td><td>3</td><td>878</td><td>68.46</td></tr><tr><td>4</td><td>urgent</td><td>4</td><td>288</td><td>25.11</td></tr></tbody></table>		priority_name character varying (10)	priority_rank smallint	responded_tickets bigint	avg_first_response_min numeric	1	low	1	828	302.62	2	normal	2	1857	149.50	3	high	3	878	68.46	4	urgent	4	288	25.11
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Q7 – Resolution by Queue	Which queues have longer resolution times?	FR-05	Compare average resolution time across queues.	<pre>SELECT q.queue_name, COUNT(f.resolution_min) AS resolved_tickets, ROUND(AVG(f.resolution_min),2) AS avg_resolution_min FROM fact_tickets f JOIN dim_queue q ON f.queue_key = q.queue_key WHERE f.resolution_min IS NOT NULL GROUP BY q.queue_name ORDER BY avg_resolution_min DESC;</pre>	Query Query History <pre>1 SELECT 2 q.queue_name, 3 COUNT(f.resolution_min) AS resolved_tickets, 4 ROUND(AVG(f.resolution_min), 2) AS avg_resolution_min 5 FROM 6 fact_tickets f 7 JOIN dim_queue q ON f.queue_key = q.queue_key 8 WHERE 9 f.resolution_min IS NOT NULL 10 GROUP BY 11 q.queue_name 12 ORDER BY 13 avg_resolution_min DESC;</pre> Data Output Messages Notifications Showing rows: 1 to 5 Page No: 1 of 1 <table><thead><tr><th></th><th>queue_name character varying (30)</th><th>resolved_tickets bigint</th><th>avg_resolution_min numeric</th></tr></thead><tbody><tr><td>1</td><td>integrations</td><td>479</td><td>5732.43</td></tr><tr><td>2</td><td>technical</td><td>1178</td><td>4981.86</td></tr><tr><td>3</td><td>onboarding</td><td>569</td><td>2248.14</td></tr><tr><td>4</td><td>billing</td><td>922</td><td>2233.21</td></tr><tr><td>5</td><td>account</td><td>627</td><td>2222.96</td></tr></tbody></table>		queue_name character varying (30)	resolved_tickets bigint	avg_resolution_min numeric	1	integrations	479	5732.43	2	technical	1178	4981.86	3	onboarding	569	2248.14	4	billing	922	2233.21	5	account	627	2222.96	
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Q8 – Resolution by Priority	Which priorities require longer resolution times?	BQ-05, FR-05	Compare average resolution time across priority levels.	<pre>SELECT p.priority_name, p.priority_rank, COUNT(f.resolution_min) AS resolved_tickets, ROUND(AVG(f.resolution_min),2) AS avg_resolution_min FROM fact_tickets f JOIN dim_priority p ON f.priority_key = p.priority_key WHERE f.resolution_min IS NOT NULL GROUP BY p.priority_name, p.priority_rank ORDER BY p.priority_rank;</pre>	Query Query History <pre>1 SELECT 2 p.priority_name, 3 p.priority_rank, 4 COUNT(f.resolution_min) AS resolved_tickets, 5 ROUND(AVG(f.resolution_min), 2) AS avg_resolution_min 6 FROM 7 fact_tickets f 8 JOIN dim_priority p ON f.priority_key = p.priority_key 9 WHERE 10 f.resolution_min IS NOT NULL 11 GROUP BY 12 p.priority_name, 13 p.priority_rank 14 ORDER BY 15 p.priority_rank;</pre> Data Output Messages Notifications Showing rows: 1 to 4 Page No: 1 of 1 <table><thead><tr><th>priority_name</th><th>priority_rank</th><th>resolved_tickets</th><th>avg_resolution_min</th></tr><tr><td>character varying (10)</td><td>smallint</td><td>bigint</td><td>numeric</td></tr></thead><tbody><tr><td>low</td><td>1</td><td>818</td><td>6544.31</td></tr><tr><td>normal</td><td>2</td><td>1822</td><td>3570.02</td></tr><tr><td>high</td><td>3</td><td>851</td><td>1568.84</td></tr><tr><td>urgent</td><td>4</td><td>284</td><td>540.74</td></tr></tbody></table>	priority_name	priority_rank	resolved_tickets	avg_resolution_min	character varying (10)	smallint	bigint	numeric	low	1	818	6544.31	normal	2	1822	3570.02	high	3	851	1568.84	urgent	4	284	540.74
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Q9 – FRT vs CSAT	How does first-response time relate to CSAT?	BQ-06, FR-09	Examine the observed association between response speed and customer satisfaction.	<pre>SELECT CASE WHEN first_response_min < 30 THEN '< 30 min' WHEN first_response_min < 60 THEN '30-59 min' WHEN first_response_min < 120 THEN '60-119 min' ELSE '120+ min' END AS response_time_group, COUNT(*) AS ticket_volume, ROUND(AVG(csat_score),2) AS avg_csat FROM fact_tickets WHERE first_response_min IS NOT NULL AND csat_score IS NOT NULL GROUP BY CASE WHEN first_response_min < 30 THEN '< 30 min' WHEN first_response_min < 60 THEN '30-59 min' WHEN first_response_min < 120 THEN '60-119 min' ELSE '120+ min' END ORDER BY MIN(first_response_min);</pre>	 <pre>1 SELECT 2 CASE 3 WHEN first_response_min < 30 THEN '< 30 min' 4 WHEN first_response_min < 60 THEN '30-59 min' 5 WHEN first_response_min < 120 THEN '60-119 min' 6 ELSE '120+ min' 7 END AS response_time_group, 8 COUNT(*) AS ticket_volume, 9 ROUND(AVG(csat_score), 2) AS avg_csat 10 FROM 11 fact_tickets 12 WHERE 13 first_response_min IS NOT NULL 14 AND csat_score IS NOT NULL 15 GROUP BY 1 16 ORDER BY 17 MIN(first_response_min);</pre> <table><tr><th>response_time_group</th><th>ticket_volume</th><th>avg_csat</th></tr><tr><td>< 30 min</td><td>197</td><td>4.03</td></tr><tr><td>30-59 min</td><td>235</td><td>4.00</td></tr><tr><td>60-119 min</td><td>286</td><td>3.90</td></tr><tr><td>120+ min</td><td>403</td><td>4.00</td></tr></table>	response_time_group	ticket_volume	avg_csat	< 30 min	197	4.03	30-59 min	235	4.00	60-119 min	286	3.90	120+ min	403	4.00
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Q10 – CSAT Distribution	What is the distribution of CSAT scores?	FR-08	Show the number and percentage of tickets receiving each CSAT score from 1–5.	<pre>SELECT csat_score, COUNT(*) AS ticket_volume, ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (),2) AS percentage FROM fact_tickets WHERE csat_score IS NOT NULL GROUP BY csat_score ORDER BY csat_score;</pre>	
Q11 – Average CSAT by Queue	What is the average CSAT by queue?	FR-08	Compare customer satisfaction across support queues.	<pre>SELECT q.queue_name, COUNT(f.csat_score) AS tickets_with_csat, ROUND(AVG(f.csat_score),2) AS avg_csat FROM fact_tickets f JOIN dim_queue q ON f.queue_key = q.queue_key WHERE f.csat_score IS NOT NULL GROUP BY q.queue_name ORDER BY avg_csat DESC;</pre>	

Q12 – Average CSAT by Priority	What is the average CSAT by priority?	FR-08	Compare customer satisfaction across priority levels.	<pre> SELECT p.priority_name, p.priority_rank, COUNT(f.csat_score) AS tickets_with_csat, ROUND(AVG(f.csat_score),2) AS avg_csat FROM fact_tickets f JOIN dim_priority p ON f.priority_key = p.priority_key WHERE f.csat_score IS NOT NULL GROUP BY p.priority_name, p.priority_rank ORDER BY p.priority_rank; </pre>	
Q13 – Overall Reopened Tickets	How many tickets are reopened?	BQ-07, FR-07	Measure the total number and percentage of reopened tickets.	<pre> SELECT COUNT(*) AS total_tickets, SUM(CASE WHEN reopened = TRUE THEN 1 ELSE 0 END) AS reopened_tickets, ROUND(100.0 * SUM(CASE WHEN reopened = TRUE THEN 1 ELSE 0 END) / COUNT(*),2) AS reopened_rate_pct FROM fact_tickets; </pre>	

Q14 – Reopened Tickets by Queue	Where do reopened tickets concentrate?	BQ-07, FR-07	Identify queues with higher reopened-ticket rates.	<pre>SELECT q.queue_name, COUNT(*) AS total_tickets, SUM(CASE WHEN f.reopened = TRUE THEN 1 ELSE 0 END) AS reopened_tickets, ROUND(100.0 * SUM(CASE WHEN f.reopened = TRUE THEN 1 ELSE 0 END) / COUNT(*),2) AS reopened_rate_pct FROM fact_tickets f JOIN dim_queue q ON f.queue_key = q.queue_key GROUP BY q.queue_name ORDER BY reopened_rate_pct DESC;</pre>	
Q15 – Open Tickets by Queue & Priority	How many tickets remain unresolved, and where?	BQ-08, FR-06	Identify open tickets and their distribution across queues and priorities.	<pre>SELECT q.queue_name, p.priority_name, p.priority_rank, COUNT(*) AS open_tickets FROM fact_tickets f JOIN dim_queue q ON f.queue_key = q.queue_key JOIN dim_priority p ON f.priority_key = p.priority_key WHERE f.is_open = TRUE GROUP BY q.queue_name, p.priority_name, p.priority_rank ORDER BY open_tickets DESC;</pre>	